

Functions of Several Variables

In Calc I and II, every function had **one input** and **one output**: $y = f(x)$.

But in the real world, most quantities depend on **more than one variable**:

- Temperature in a room depends on position (x, y, z)
- Your grade depends on homework, quizzes, and exams
- Wind chill depends on temperature *and* wind speed

Function of Two Variables

A **function of two variables** is a rule that assigns to each ordered pair (x, y) in a set $D \subseteq \mathbb{R}^2$ a unique real number z :

$$z = f(x, y)$$

- **Domain**: the set D of valid inputs (x, y)
- **Range**: the set of all output values $z = f(x, y)$

This extends naturally: $w = f(x, y, z)$ is a function of **three** variables, and so on.

Same idea as Calc I, but now the input is a **point** in $\mathbb{R}^2$ (or $\mathbb{R}^3$) instead of a single number.

Example 1: Finding the Domain

Example 1

Find and sketch the domain of $f(x, y) = x \ln(x^2 - y)$.

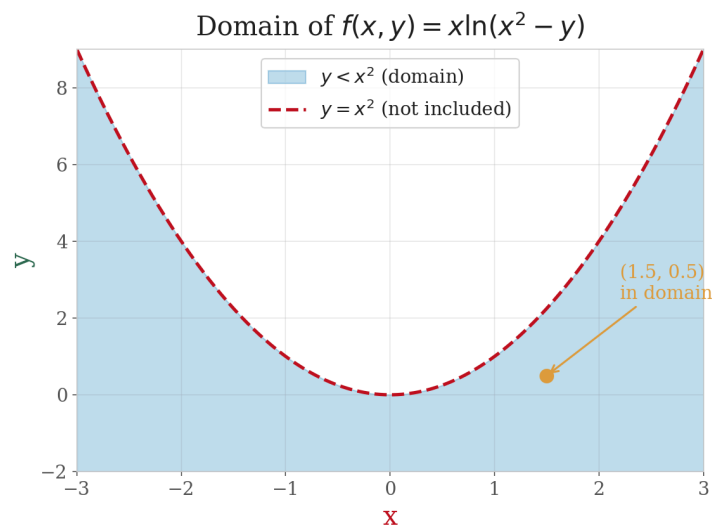
Look for restrictions, set up and solve inequalities, sketch.

We need $x^2 - y > 0$, which means $y < x^2$.

The domain is the set of all points (x, y) that lie **strictly below** the parabola $y = x^2$.

$$D = \{(x, y) \mid y < x^2\}$$

The boundary $y = x^2$ is **not** included (we need strictly greater than zero inside the logarithm).



Domain: the region below $y = x^2$

Your Turn: Find the Domain

Example 2

Find the domain of $g(x, y) = \frac{\sqrt{y - x^2}}{1 - x^2}$.

Identify both restrictions, then describe and sketch the domain.

Restriction 1: Square root requires $y - x^2 \geq 0$, so $y \geq x^2$.

Restriction 2: Denominator requires $1 - x^2 \neq 0$, so $x \neq \pm 1$.

$$D = \{(x, y) \mid y \geq x^2 \text{ and } x \neq \pm 1\}$$

This is the region **on or above** the parabola $y = x^2$, with two vertical lines removed.

Graphs of Functions of Two Variables

In Calc I, the graph of $y = f(x)$ was a **curve** in $\mathbb{R}^2$.

Now the graph of $z = f(x, y)$ is a **surface** in $\mathbb{R}^3$:

Graph of a Function of Two Variables

The **graph** of f is the set of all points (x, y, z) in $\mathbb{R}^3$ such that $z = f(x, y)$ and $(x, y) \in D$.

$$\text{graph} = \{(x, y, z) \mid z = f(x, y), (x, y) \in D\}$$

Input = point in the xy -plane. Output z = height above (or below) that point. The collection of all these heights forms a **surface**.

Example 3

Sketch the graph of $f(x, y) = -3x - 2y + 6$.

Recognize: This is $z = -3x - 2y + 6$, which is **linear** in x and y . A linear equation in three variables is a **plane**.

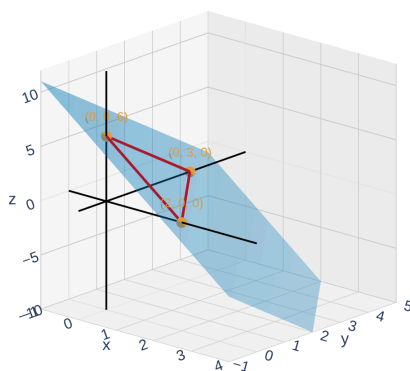
Find the intercepts and sketch the plane.

Find the intercepts by setting two variables to zero:

- x -intercept: set $y = 0, z = 0 \rightarrow x = 2 \rightarrow$ point $(2, 0, 0)$
- y -intercept: set $x = 0, z = 0 \rightarrow y = 3 \rightarrow$ point $(0, 3, 0)$
- z -intercept: set $x = 0, y = 0 \rightarrow z = 6 \rightarrow$ point $(0, 0, 6)$

Connect these three points to form a triangular portion of the plane in the first octant. The full plane extends in all directions.

$$z = -3x - 2y + 6$$



The plane $z = -3x - 2y + 6$ | [Interactive version](#)

Traces: Slicing a Surface

A 3D surface can be hard to visualize all at once. One strategy: **slice it** with a plane and look at the resulting curve.

Traces

A **trace** of a surface is the curve of intersection when the surface is sliced by a plane.

- **Horizontal trace** (or *cross-section*): set $z = c$ (a constant) and see what curve remains in the xy -plane
- **Vertical trace**: set $x = c$ or $y = c$ and see the resulting curve

Analogy: Think of slicing a loaf of bread. Each slice reveals a 2D cross-section of the 3D shape.

Or think of a topographic map — each contour line is where the terrain has a specific elevation.

Traces reduce a 3D problem to a **2D curve** — something you already know how to analyze from Calc I.

Level Curves and Contour Maps

Level Curves

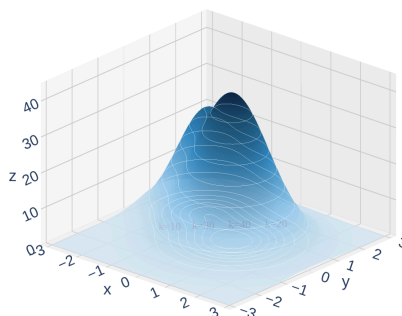
The **level curves** of $f(x, y)$ are the horizontal traces $f(x, y) = k$ for various constants k .

Each level curve is drawn in the xy -plane. Viewed together, they form a **contour map** (or *contour plot*).

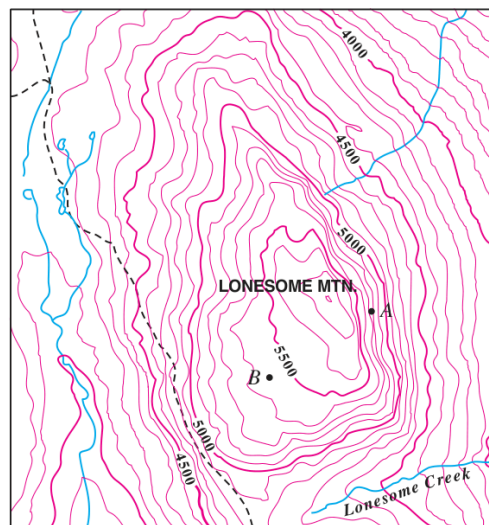
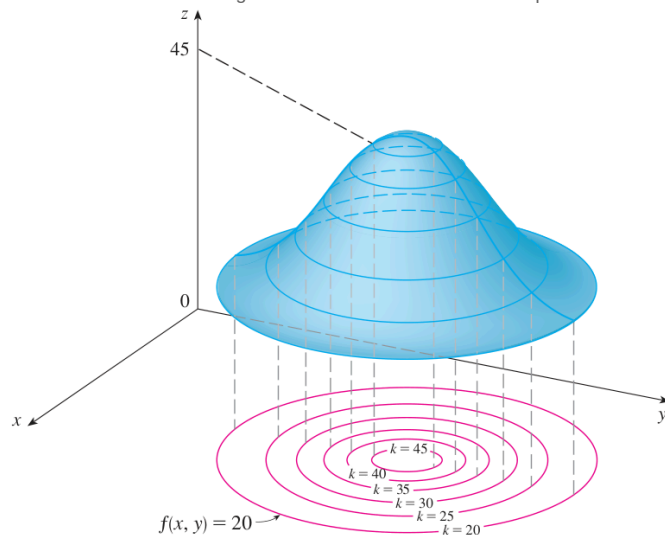
You've seen these before!

- Topographic maps: curves of constant elevation
- Weather maps: curves of constant temperature (isotherms) or constant pressure (isobars)
- Closely spaced contour lines = the surface is **steep**
- Widely spaced = the surface is **gentle**

Surface with Level Curves Projected Below



Drag to rotate --- see how the surface produces the contour map on the floor | [Interactive version](#)



Contour lines show constant elevation

A contour map compresses 3D information into a 2D picture. Reading one is like reading a topographic map — each curve is a constant height $z = k$.

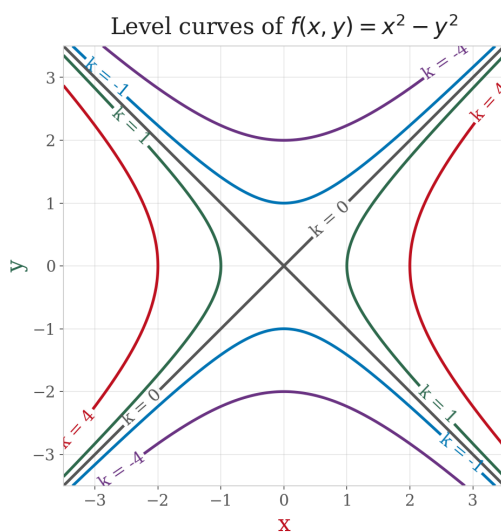
Example 4: Contour Plot of $f(x, y) = x^2 - y^2$

Example 4

Sketch the level curves of $f(x, y) = x^2 - y^2$ for $k = -4, -1, 0, 1, 4$.

- $k = 0$: $x^2 - y^2 = 0 \rightarrow y = \pm x$ (two lines through the origin)
- $k > 0$ (e.g., $k = 1, 4$): $x^2 - y^2 = k \rightarrow$ **hyperbolas** opening left and right
- $k < 0$ (e.g., $k = -1, -4$): $x^2 - y^2 = k \rightarrow$ **hyperbolas** opening up and down

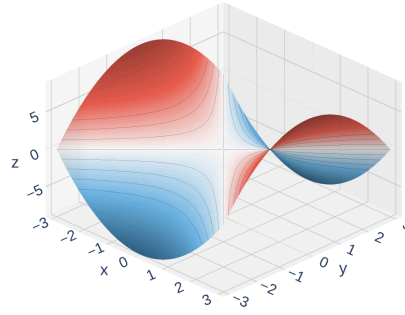
The surface $z = x^2 - y^2$ is a **hyperbolic paraboloid** (saddle surface). The contour map reveals this saddle shape through its family of hyperbolas rotating through the lines $y = \pm x$ at $k = 0$.



Level curves of the hyperbolic paraboloid $z = x^2 - y^2$

Here is the saddle surface with level curves drawn on it — rotate to a top-down view to see the contour map:

$$z = x^2 - y^2 \text{ (hyperbolic paraboloid)}$$

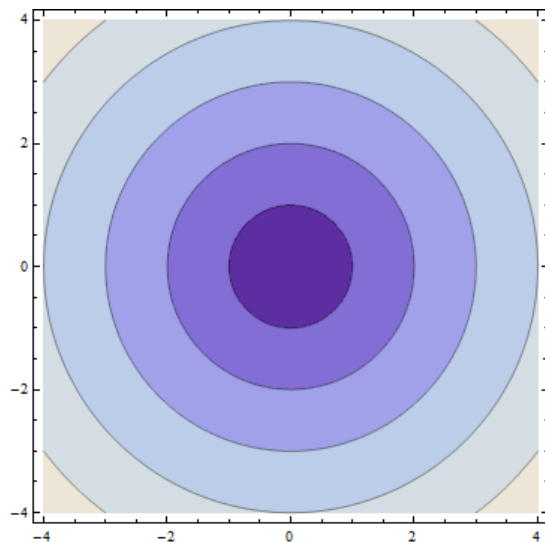
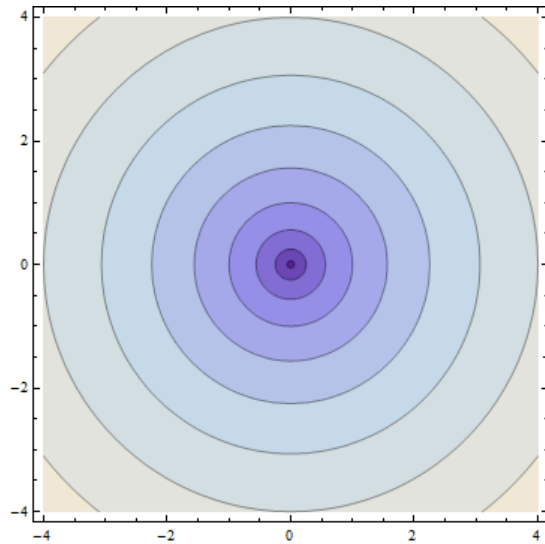


Drag to rotate --- try looking straight down! | [Interactive version](#)

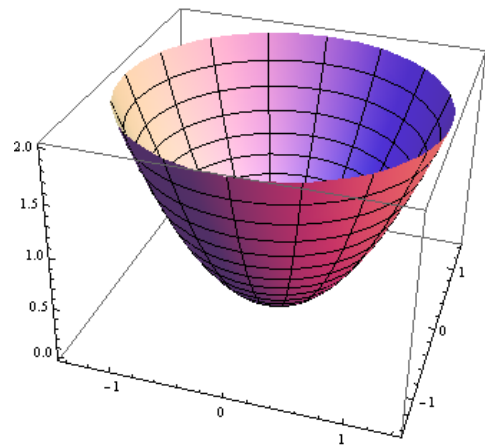
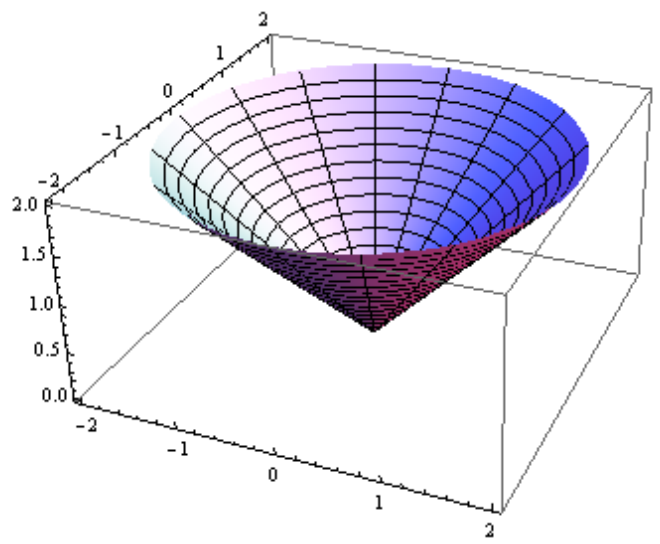
Match the Contour Plot to the Surface

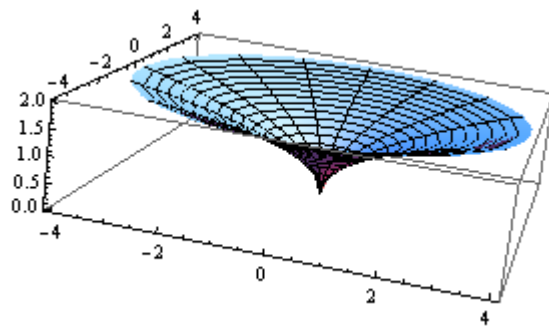
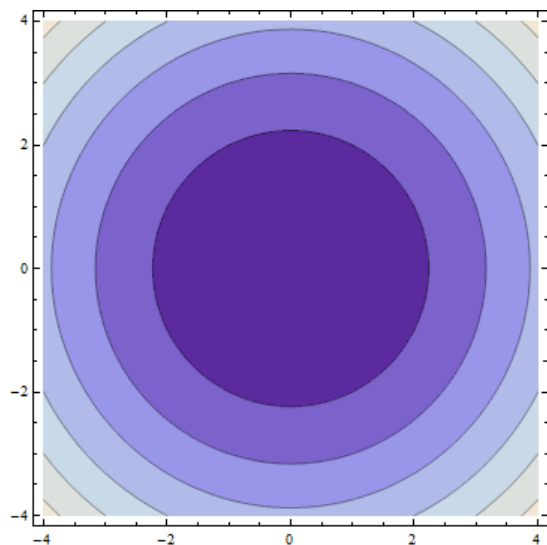
Challenge: Can you match each contour plot (left) to its surface (right)?
Look for clues: symmetry, shape of level curves, spacing of contour lines.

Contour Plots



Surfaces



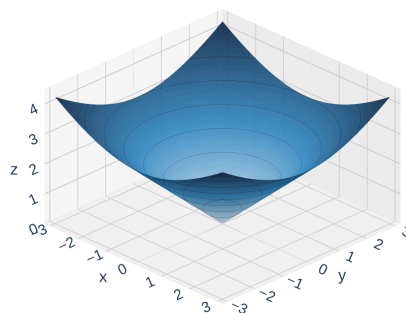


Write your matches and explain your reasoning.

The Equations Behind the Surfaces

All three surfaces have the form $z = r^p$ where $r = \sqrt{x^2 + y^2}$. The **power** p controls the contour spacing. Rotate to look straight down to see the contour map!

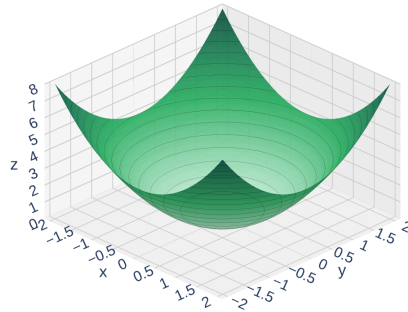
Cone: $z = \sqrt{x^2 + y^2}$ — constant slope, so contours are **evenly spaced**.
 $z = \sqrt{x^2 + y^2}$



[Interactive version](#)

Paraboloid: $z = x^2 + y^2$ — slope increases outward, so contours get **closer together**.

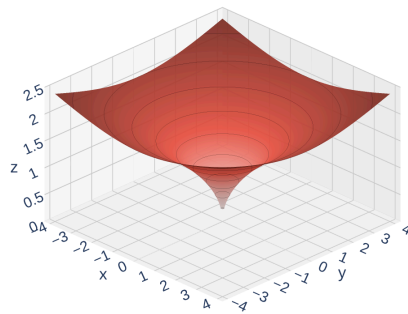
$$z = x^2 + y^2$$



[Interactive version](#)

Fourth root: $z = (x^2 + y^2)^{1/4}$ — slope decreases outward, so contours get **farther apart**.

$$z = (x^2 + y^2)^{1/4}$$



[Interactive version](#)

Level Surfaces: Functions of Three Variables

What if the function has **three** input variables, like $w = f(x, y, z)$?

We can't graph $w = f(x, y, z)$ directly — that would require **4 dimensions**!

Instead, we visualize using **level surfaces**: set $f(x, y, z) = k$ for various constants k .

Each value of k gives a **surface** in $\mathbb{R}^3$ (not just a curve).

Level Surfaces

The **level surfaces** of $f(x, y, z)$ are the surfaces defined by $f(x, y, z) = k$ for constants k in the range of f .

Analogy:

- For $f(x, y)$: level *curves* (1D objects in 2D space)
- For $f(x, y, z)$: level *surfaces* (2D objects in 3D space)
- Same idea, one dimension up!

Level curves collapse a 3D graph to 2D. Level surfaces collapse a 4D graph to 3D. Both let us visualize functions we can't directly graph.

Example 6: Level Surfaces of a Temperature Field

Example 6

A temperature distribution is given by:

$$T(x, y, z) = \frac{100}{(x^2 + y^2 + z^2)^{3/2}}$$

Describe the level surfaces. What shape are the level surfaces? What happens to temperature as you move away from the origin?

Set $T = k$ and solve for the relationship between x , y , and z .

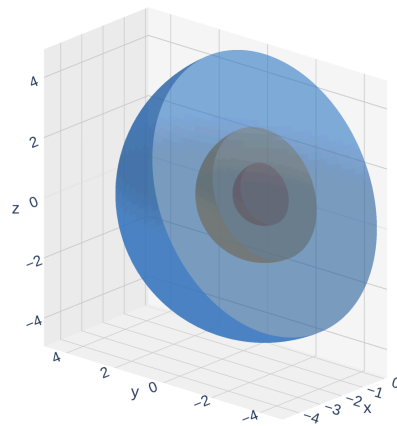
Set $T = k$:

$$\begin{aligned}\frac{100}{(x^2 + y^2 + z^2)^{3/2}} &= k \\ (x^2 + y^2 + z^2)^{3/2} &= \frac{100}{k} \\ x^2 + y^2 + z^2 &= \left(\frac{100}{k}\right)^{2/3}\end{aligned}$$

This is a **sphere** of radius $r = \left(\frac{100}{k}\right)^{1/3}$ centered at the origin.

- Larger k (higher temperature) $\rightarrow$ smaller radius $\rightarrow$ closer to the origin
- Temperature is **hottest at the origin** and decreases as you move outward
- The level surfaces are **concentric spheres** (like layers of an onion)

Level surfaces: $T = 100 / (x^2 + y^2 + z^2)^{3/2}$



Hemispheres cut at $x = 0$ to show nesting | [Interactive version](#)

Homework

Section 14.1: 3, 9, 13, 23-31, 32, 36, 41, 42, 45, 47, 49, 61-66, 67, 68